Fast Power Cycling Test for IGBT Modules in Traction Application

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Abstract The numerous advantages of Insulated Gate Bipolar Transistor (IGBT) power modules and their ongoing development for higher voltage and current ratings make them interesting for traction applications. These applications imply high reliability requirements. One important requirement is the ability to withstand power cycles. Power cycles cause temperature changes which lead to a mechanical stress that can result in a failure. Lifting of bond wires is thereby the predominant failure mechanism.
A fast power cycling test method activating the main failure mechanism has been developed which allows to reproduce millions of temperature changes in a short time. The applicability of fast testing is supported by a mechanical analysis. Test results show the number of cycles to failure as a function of temperature changes for an IGBT single switch. A descriptive model is deduced from the results.

1. Introduction

IGBT modules have become important devices for static power conversion, variable speed drives and uninterruptable power supplies. They have replaced Bipolar Transistors and MOSFETs in many applications due to a suitable combination of a frequency range up to 20kHz, low on state resistance, a simple gate drive and blocking voltages up to 3kV. The further advances of IGBT technology for higher voltages and currents will lead to a competition with traditional press packed devices in the high current/voltage range such as thyristors and GTOs [4].
The above features make IGBT modules interesting for traction applications such as drives for busses, trams, trains, elevators and electrical cars. Reliability is of the greatest importance in traction applications. Resulting from the service profile of a traction vehicle one important reliability criteria is the capability to withstand up to 10 million *power cycles* defined as the alternating change of high and zero (or low) power consumption and corresponding to the motion and stopover of the vehicle. Power cycles impose a rise and fall of temperature (ΔT) and therefore a cyclic thermomechanical stress which can lead to a failure of the module. Lifting of bond wires has been identified as the main failure mechanism caused by power cycles [1,2].
Basically two test methods are used to determine power cycle withstand capability:
a) *active power cycles* rebuild the ΔT and the timing of real power cycles by means of powering the module periodically.
b) *passive temperature cycles* , e.g. the module is externally heated and cooled in order to achieve the desired ΔT.
These two methods suffer from extremely long test times since millions of power cycles are expected in traction applications. Acceleration of both methods by increasing ΔT is controversal due to the activation of different material related mechanisms [7].
The objective of the investigation was to develop a time-saving active power cycling method which imposes a stress on the bond connection with realistic ΔT and therefore activates the main failure mechanism of bond wire lifting.

2. Reliability Requirements

Reliability, e.g. the ability to perform a required function under given conditions for a stated time interval for power modules in traction application means, in simple terms, switching under defined electrical parameters under the expected environmental conditions during the useful life. The reliability of a module is affected by a variety of application specific factors such as electrical load and environmental conditions (temperature, humidity, vibration, mechanical shocks etc). Special attention in traction is payed to temperature and temperature changes which can arise from: type 1) seasonal influences winter/summer, type 2) daily changes night/noon and on duty/off duty, type 3) changes due to long stopovers at terminal stations and type 4) power cycles. Power cycles are considered to be the most severe due to their large number and high temperature swing.

Table 1 Expected lifetime temperature changes ΔT and number of cycles in urban railway applications

	type 1	type 2	type 3	type 4
ΔT [K]	20-40	10-30	30-80	30-80
# cycles	$<10^2$	$<10^4$	$<10^5$	10^6-10^8

As an example the number of application power cycles of a tram is projected for the following service profile:

Expected life time of tram:	30 years
Service time in 30 years: 30 years × 300 days/year × 15h/day	135'000h
Motion time between stopover:	30s
Time at stopover:	30s
Number of power cycles in 30 years:	8'100'000

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3. Module Assembly

Most module types consist of three main parts: a DBC (Direct Bonded Copper) substrate carrying chips and connections, a thick copper base plate and a plastic housing. Humidity protection is achieved by a soft silicon gel (S) sometimes in addition with a hard epoxy resin (H). Figure 1 and Table 2 give an overview of typical measures and material properties of an IGBT module.

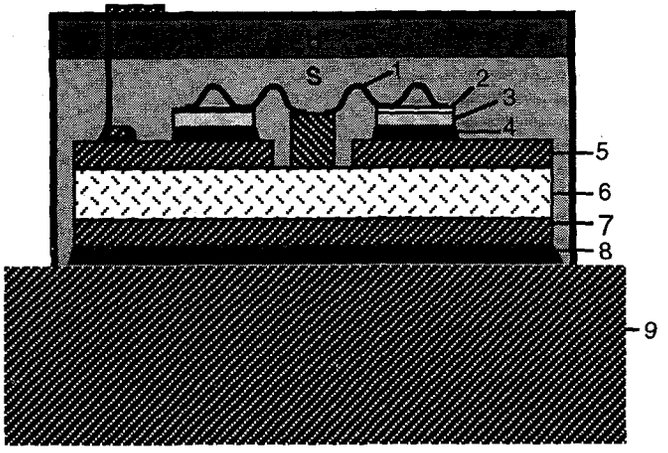


Fig.1 Basic crosssectional view of a power module assembly (not to scale)

Table 2 Typical measures and relevant data of power module materials

	Material	Thickness [µm]	$\lambda_{20°C}$ [W/mK]	$\alpha_{20°C}$ [10^{-6}/K]
1) Bond wire 2) Chip metall	Al	300 3	≈240	24
3) Chip	Si	300	80-150	3
4) Die attach 8) DCB attach	Solder	50 100	≈35	≈25
5) DCB upper layer 7) DCB lower layer 9) Mounting plate	Cu	300 300 3000	390	16.8
6) DCB ceramic	Al_2O_3 AlN	700	≈35 140-170	≈7 ≈4

IGBT chips are soldered onto the DBC substrate. Several chips are connected in parallel in order to achieve a desired current capability. The structured DBC layers contact via solder joints or bond wires the collector, emitter and gate of the transistor chip. The ceramic fullfills two main tasks: electrical isolation of conducting structures and heat transfer towards the mounting plate. Aluminumoxide Al_2O_3 and Aluminum-Nitride AlN are mainly used the latter being advantageous concerning thermal conductivity. A prebent robust copper plate is used as heat spreader and in order to mount the module onto an air or water cooled heat sink.

The metallization of the chip representing the emitter and containing the gate of the IGBT is contacted by means of wedge bonds. The bond wires consist of 4-nine or 5-nine (99.99, 99.999%) aluminum with small amounts of titanium, copper or other. These alloy components and the wire pull process determine the grain size and mechanical properties of the wire. The aluminum of the chip metallization consists generally of 99% aluminum plus 1% of silicon or titanium.

4. Failure mechanisms and indicator

Failure mechanisms caused by power cycles are package related: 1) lifting of bond wires, and 2) solder joint fatigue [1]. Bond wire lifting is regarded as the dominant failure mechanism caused by power cycles because it occurs before thermal fatigue of solder joints [1,2,7].
The lifting of a bond wire can be regarded as a domino effect because it leads to an non-homogeneous current distribution on the IGBT chip and therefore to higher local and average temperatures which accelerate the lifting of further wires.
Lifting of bond wires is caused by crack growth which is induced by the thermomechanical stress caused by the rise and fall of temperature and the different coefficients of thermal expansion CTE of silicon and aluminum.

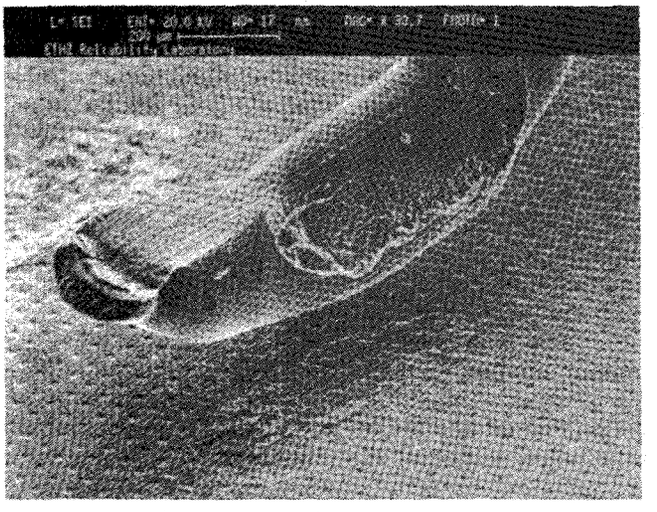


Fig. 2 SEM image of a lifted bond wire

Cracks grow along the boundary of fine and coarse aluminum grains inside the bond wire from the heel and toe of the bond foot to the center [2,3]. It is assumed that the difference in grain size is caused by the wegde bond process. When a crack reaches the center the bond wire lifts i.e. the mechanical and electrical contact to the metallization is interrupted. Consequently the remaining wires have to conduct more current causing a current redistribution in the chip and an increased average temperature.

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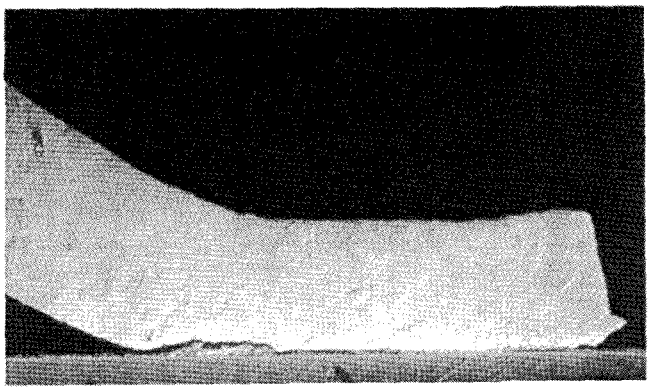


Fig.3 Cross-sectional view of a crack in a bond wire

Another effect of the thermomechanical stress is reconstruction, e.g. the extrusion of aluminum grains of the chip metallization [6]. As a consequence the sheet resistance increases and therefore a higher VCE at constant collector current IC results. This leads to an increase in power dissipation and temperature and therefore again stress. A second effect of severe reconstruction is the interruption of the metallization in single emitter windows and therefore a disconnection of IGBT cells. Eventually the conducting area decreases dramatically which leads to current densities that cause melting or electromigration-like effects.

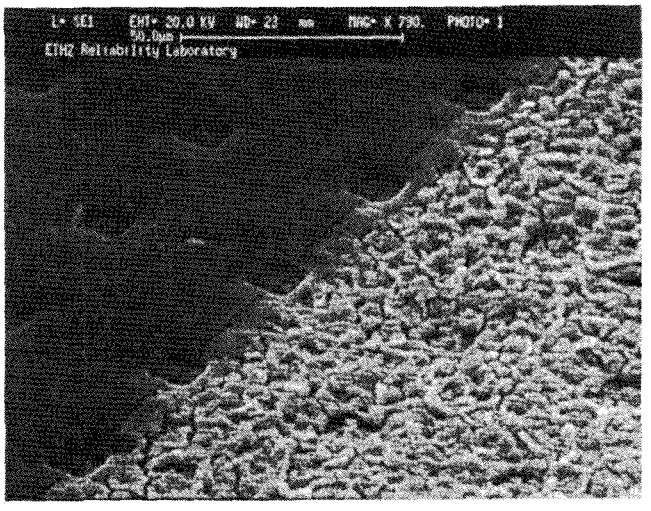


Fig.4 SEM image of reconstruction of chip metallization on non passivated areas

Remarkably, reconstruction does not occur in areas of the metallization which are covered by a passivation layer [6]. However, the bond contact windows cannot be passivated and take up a large area - about 50% - of the metallization.
Indicator for bond wire lifting and reconstruction is an increase of the collector-emitter voltage VCE. This is due to the fact that both mechanisms cause a rise of the average chip temperature. For the investigated IGBT technology VCE rises with temperature. Therefore VCE can be used as an indicator for the course of both related failure mechanisms.

VCE can be measured by two methods: 1) online, i.e. during the on time of the power cycle, or 2) static, which means that the power cycle test is interrupted periodically for VCEsat measurement at defined conditions at thermal steady state. Static measurement is preferred as reliability indicator because it allows the comparison of modules that are tested under different electrical and timing conditions. The current for static VCEsat measurement is the rated current of the module.

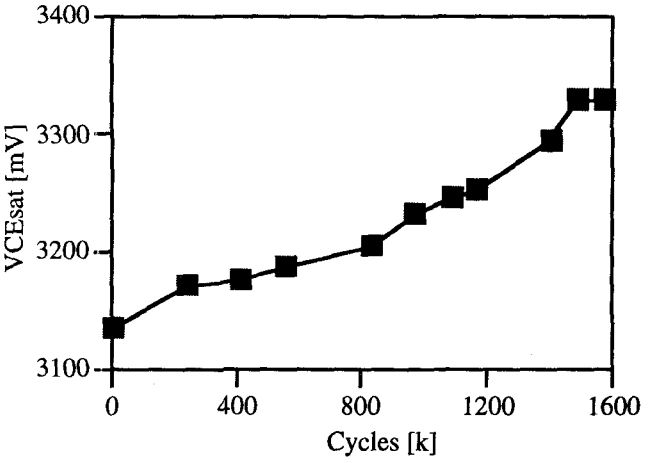


Fig.5 VCEsat as function of power cycles, static measurement

The percentage of VCE increase that is critical and therefore used as failure criteria depends on the rated current of the module. As a general rule for modules with several 75A IGBT chips in parallel the failure criteria is defined as:

$$\Delta VCE[\%] = \frac{1500}{Irated[A]} \qquad \text{Eq. 1}$$

The failure criteria for a 300A module for example is then an increase of VCE of 5%.

4. Fast power cycle testing

Power cycle withstand capability as an important part of reliability requirements in traction application must be determined. Because a one to one test of several million power cycles at a cycle period of about one minute would last for years a faster test method has been developed.
Time-saving for the investigation of the main failure mechanism - lifting of bond wires - can be achieved by applying shorter cycle periods. This is due to the quick thermal response at the critical connection of bond wire and chip metallization located close to the heat generating source within the IGBT chip. This can be easily veryfied by calculation or measurement. Figure 6 shows the calculated thermal response at different locations inside an 300A/1200V single switch IGBT module.

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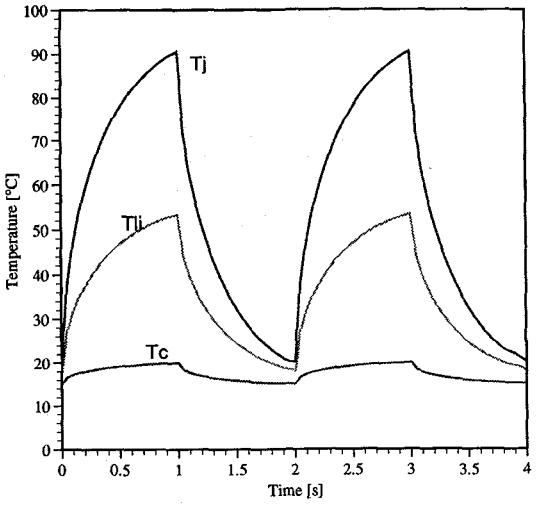


Fig. 6 Calculated thermal response

In figure 6 Tj represents the temperature in the heat generating junction of the IGBT transistors, Tli is the temperature in the ith layer of the module assembly and Tc the temperature of the mounting plate.
In practice power cycle tests are carried out with the IGBT modules mounted on a watercooled heatsink. A specified load current Iload is periodically applied to the IGBT whose gate is permanently set to a constant voltage. The gate voltage V_{GE} thereby must be set to a value significantly above the gate-emitter threshold voltage $V_{GE(th)}$ in order to assure a homogeneous current distribution among the IGBT cells in a chip and among paralleled chips.

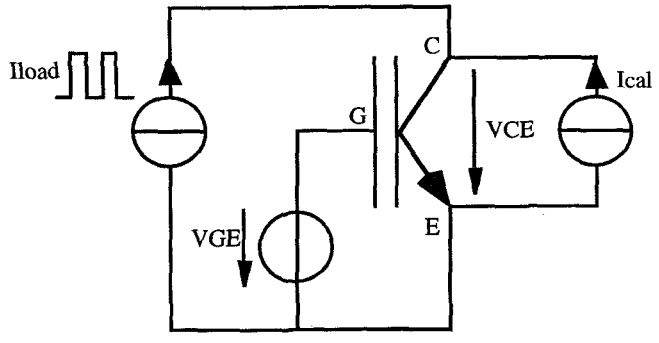


Fig. 7 Principle circuit of power cycling

The main driving parameters for the bond connection failure mechanism are the temperatures at the bond contact and the chip metallization which due to the proximity of these locations to the heat generating chip are approximately the junction temperature Tj. Therefore Tjlow and Tjhigh are the parameters to be set initially in power cycle tests by means of adjusting Iload, ton, toff and the cooling to appropriate values. Tjlow, Tjhigh and the resulting VCE are not further controlled during the test, thus an increase of VCE will lead to a shift of the junction temperatures. However, the variability of VCE is desireable because it corresponds to the real application.

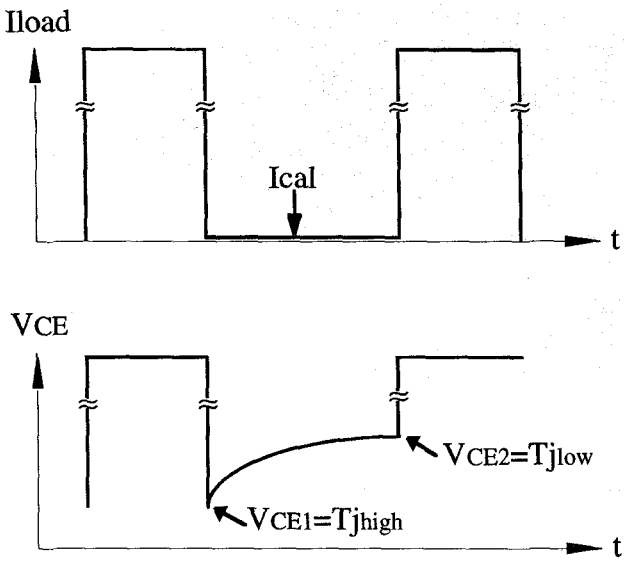


Fig. 8 Principle of Tj determination

The junction temperatures Tjlow and Tjhigh are measured by means of a TSEP (Temperature Sensitive Electrical Parameter). For IGBTs the collector-emitter voltage V_{CE} caused by a small calibration current Ical (e.g. 100mA) is the most suitable TSEP. For the investigated IGBTs the relation between junction temperature and V_{CE} with constant Ical is linear and inversely proportional. Using the previously measured calibration characteristics the junction temperatures can be easily determined by applying Ical during the off time of a power cycle.

5. Mechanical analysis

The following analysis of mechanical stress assumes a simplified two-layer structure consisting of silicon and aluminum in which presumably all plastic deformation takes place in the aluminum due to its comparatively low yield strength. The equivalent strain amplitude caused by a power cycle depends on its ΔT and the $\Delta\alpha$ of the two materials [2]:

$$\Delta\varepsilon_{eq} = c \cdot \Delta T \cdot \Delta\alpha \qquad \text{Eq.2}$$

where c is a constraint factor taking into account that strain is present in both in-plane directions. c is expressed as:

$$c = 1/(1-\nu) \qquad \text{Eq.3}$$

with the poisson ratio ν for aluminum being 0.345 in the elastic range and approaching 0.5 in the fully plastic range. The stress increment per K for Al is then:

$$\frac{\sigma}{\Delta T} = \Delta\alpha \cdot E \cdot c \qquad \text{Eq.4}$$

which is 2.3 MPa/K for elastic deformation using the values of table 3.

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Table 3 Material properties

	Al	Si
CTE α [10^{-6}/K]	24	3
Young's modulus E [GPa]	72	130
Yield strength σ_y [MPa] at εpl=0.2%	20	-
Poisson ratio ν [–]	0.345	0.23

Plastic deformation is therefore already reached with a ΔT of 9K provided a yield strength σ_y of 20MPa for the 99.99% aluminum bond wire. Thus, with the ΔT in power cycles being larger than 18K (i.e. $2\Delta T$ to reach the yield strength $-\sigma_y$ ... $+\sigma_y$ cyclically) an instantaneous plastic deformation takes place in the aluminum being the compliant part of the connection. The plastic deformation is restricted to a layer of aluminum of several microns in thickness adjacent to the bond region and covers the entire area of the bond foot. Cyclic plastic deformation results in fatigue crack growth along the boundary of coarse and fine grains in the bond wire [2,3] which ultimately causes the lifting. However, this approach does not cover the stress and strain peaks which occur at the toe and heel of the wedge bond [2,5]. Taking such stress concentrations into account this lower bound approach described above could even underestimate the amount of deformation that actually occurs.

The yield strength σ_y of the aluminum metallization is obviously higher than that of the bond wire due to its small grains, amounts of alloying elements (Si, Ti etc.), and a considerable density of lattice defects owing to the sputtering process. This would require a higher ΔT to initiate plastic deformation of the metallization. Cyclic plastic deformation of the metallization will then cause reconstruction.

Power modules operate up to a moderate homologous temperature in the order of 0.4 for aluminum. Therefore, creep could occur in case of considerable stress. However, an exact distinction between creep and plasticity seems to be difficult in power cycling experiments with its varying temperatures and stress states. In fast power cycles creep cannot occur due to zero dwell times, i.e. at the upper turning point of temperature the stress is reduced by unloading rather than by relaxation. In order to correlate fast (periods of 0.75 to 9s) and real (periods of 60s) power cycles, the amount of possible creep is estimated.

Since the total strain introduced by the thermal mismatch is fixed, the amount of time dependent deformation, i.e. creep strain, can only convert some part of the elastic strain to reduce the stress (i.e. relaxation).

With $\varepsilon tot = \varepsilon pl + \varepsilon el = c \cdot \Delta T \cdot \Delta\alpha$ and $\Delta\varepsilon el = 2\sigma y/E$ the ratio of elastic and plastic strain amplitude $\Delta\varepsilon el / \Delta\varepsilon pl$ is:

$$\frac{\Delta\varepsilon_{el}}{\Delta\varepsilon_{pl}} = \frac{\Delta\varepsilon_{tot}}{\Delta\varepsilon_{pl}} - 1 = \frac{c \cdot \Delta T \cdot \Delta\alpha}{c \cdot \Delta T \cdot \Delta\alpha - \frac{2 \cdot \sigma_y}{E}} - 1 \qquad \text{Eq.5}$$

An exemplary calculation for ΔT=50K, the material properties of Al, and c taken as 2 for the plastic range shows that the ratio of elastic and plastic strain is approximately 30%, which in analogy represents the amount of possible creep strain in comparison to the plastic deformation. An assessment using relevant material properties has shown that the inelastic strain additionally introduced by creep is at most 1/3 of εel . Therefore, creep could add approximately 10% to the inelastic strain amplitude in the case of a sufficiently long dwell time for relaxation. Therefore, the fast power cycling test produces approximately the same damage and failure mode as slower cycling experiments.

6. Test results

Fast power cycle tests were carried out with 300A/1200V single switch IGBT modules. Three medium temperatures Tm of 60, 80 and 100°C and ΔTj of 30 to 80K were preset. Goal was to investigate the influence of Tm and ΔTj on the number of cycles to failure. The following conditions have to be set in order to achieve the above cycles:

VGE=15V, current Iload between 240 and 300A, ton = 0.6 to 4.8s, toff between 0.4 and 5s, watercooled heat sinks. Failure criteria is a 5% increase of VCE, registered by periodical static measurement.

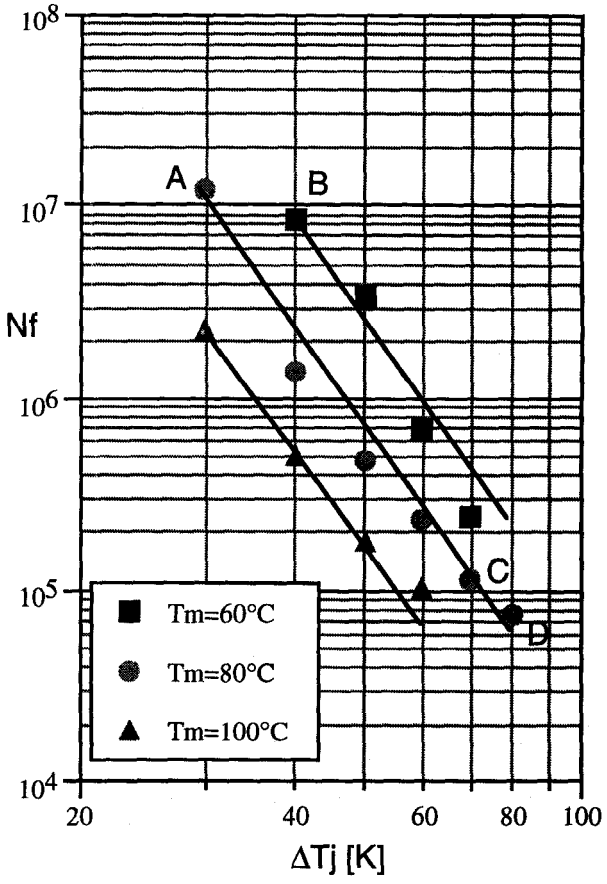


Fig. 9 Number of cycles to failure Nf as function of ΔTj with Tmedium as parameter

Displaying the results in a double logarithmic scale a dependence of Nf on the medium temperatur Tm and the temperature difference ΔTj is clearly evident.

Failure analysis showed reconstruction of the chip metallization and lifted bond wires. Qualitatively, reconstruction was mild for modules A and B, strong for C and D, and medium for all remaining modules. Lifted bond

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wires were found not to be assignable to certain Tm/ΔTj combinations. Electrical characterization revealed no change in IGES, ICES and VGE(th).
With some precaution it can be stated that the power cycle capability of the bond connection of the tested module type is close to the requirements for traction application provided a ΔTj lower than 40K and medium temperatures Tm not higher than 60°C. This could be achieved by means of a large derating of the collector current.

7. Descriptive Model

Based on the test results a general approach to model the number of cycles to failure is proposed in which ΔTj and Tm are apparently the relevant parameters. The model is descriptive only which means that it neither considers the physical structure of the module nor the actual failure mechanisms of reconstruction and crack growth. Nevertheless, the model contains some physical relevance because the plastic deformation is related to ΔTj and materials properties depend on Tm.
The straight lines in the double logarithmic scale of figure 9 imply a power law for the dependence of Nf on ΔT. An almost parallel shift for different Tm indicates a thermally activated mechanism and can therefore be expressed by an Arrhenius approach. The combination of the two approaches is then:

$$Nf = A \cdot \Delta Tj^{\alpha} \cdot \exp\left(\frac{Q}{R \cdot Tm}\right) \qquad \text{Eq.6}$$

with R being the gas constant (8.314 J/mol·K) and Tm in Kelvin make A = 640, $\alpha = -5$, and $Q = 7.8 \cdot 10^4$ $Jmol^{-1}$ (or about 0.8eV).
The validity of this descriptive model beyond the provided results, i.e. for ΔT smaller than 30K or higher than 80K is not yet determined. Moreover, it is obvious that the above approach could be restricted to investigated module type.

8. Conclusion

Fast power cycle testing is suitable to investigate the connection of metallization and bond wire. Reconstruction of metallization and crack growth in bond wires are observed in power cycle tests as failure mechanisms. Reconstruction can be completely prevented by means of a hard passivation layer. VCE is an appropriate parameter to trace the course of these mechanisms for the tested modules. A descriptive model is proposed which allows to determine the number of cycles to failure as a function of ΔTj and Tm.
However, the investigation showed that a deeper understanding of the failure mechanisms is necessary. Experiments to elucidate these effects are not finished up to now.

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